

SQL & Database Optimization

Query Tuning, Schema Design, Indexing, Sharding & High Availability

QUERY TUNING & INDEXING

1. Deep Execution Plan Analysis (EXPLAIN ANALYZE)

Diagnose sequential scans, nested loop spills, buffer misses, and cardinality misestimations.

2. Composite Indexing & Leftmost Prefix Rule

Order columns for equality vs range conditions to maximize index traversal efficiency.

3. Eliminating Non-SARGable Anti-Patterns

Refactor functions on indexed columns, implicit type casting, and wildcards into SARGable queries.

4. Pagination Optimization: Keyset vs Offset

Replace O(N) OFFSET queries with deterministic keyset/cursor pagination for multi-million row tables.

SCHEMA DESIGN & SCALE

5. PostgreSQL Declarative Table Partitioning

Implement Range, List, and Hash partitioning with automated sub-partition management via pg_partman.

6. High-Concurrency Queue Tables (SKIP LOCKED)

Build zero-contention task queues using SELECT ... FOR UPDATE SKIP LOCKED across parallel workers.

7. Zero-Downtime Database Schema Migration

Execute Expand-and-Contract patterns, dual-writing, asynchronous backfills, and online cutovers.